

Lecture 1: Course Info, Motivation from Numerical Analysis, and Stochastic Toolkit

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Introduction to Data Assimilation
2026 Spring



Outline

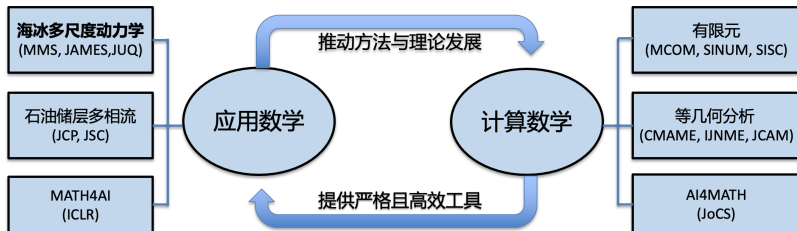
1. Course Info
2. Motivation from a numerical analysis point of view
 - Improving accuracy: error accumulation over time in time-marching schemes
 - Improving stability: conditional stability, such as CFL
3. Probability and stochastic toolkit
 - Basic concepts of probability
 - Introduction to Itô stochastic calculus, Wiener process (Brownian motion), Itô's formula/lemma

Self Introduction

- YMSC, 2025–present, assistant professor; ANU honorary lecturer
- ANU, 2022–2025, lecturer
 - Comput. Math.: **SoftFEM/SoftIGA**, fault-tolerant for adaptive FEM
 - Appl. Math.: **Ice floe multiscale modeling and data assimilation**
 - AI4Science: ML-assisted sea ice modeling, quantum many-body problems
 - Math4AI: Explainable AI (XAI) via Rashomon sets, neural networks
- UW-Madison, 2020–2022, van Veleck visiting assistant professor
 - Sea ice floe modeling, data assimilation, Kalman filters
 - **SoftFEM**, outlier removal for IGA spectrum
- More information at the YMSC page or my GitHub page
<https://quanlingdeng.github.io/>

Research Interests

- **Applied and Computational Mathematics**: sea ice floe dynamics, multiscale modeling, ocean and atmosphere dynamics, **data assimilation**, **Kalman filters**, UQ, parallel computing, PDE solvers such as FEM, IGA, HHO, generalized-alpha methods, error analysis/estimates.
- **AI/ML**: deep neural networks, CNN, U-net, PINN, object detection, feature selection, Rashomon set, quantum AI, AI for Math/Science.



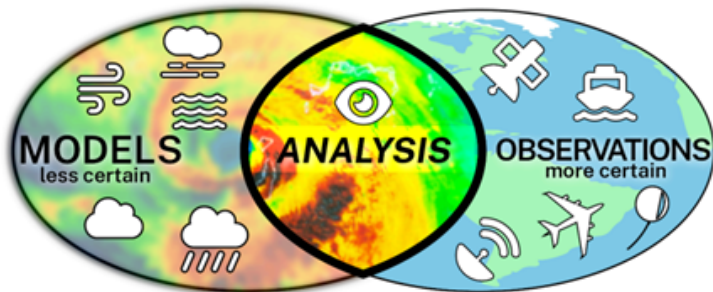
Data Assimilation

What is Data Assimilation?

Fundamental Challenge

Prediction = Model + Data

- Models are incomplete
- Observations are noisy and sparse
- Initial conditions are uncertain



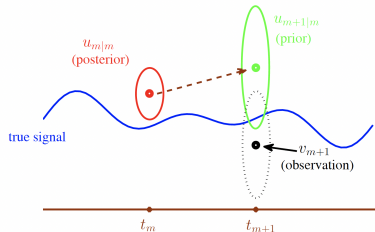
Formally: What is Data Assimilation?

Data assimilation (DA) is a mathematical framework for combining observational data with numerical models to obtain the best possible estimate of the state of a physical system.

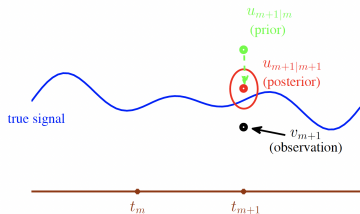
Model **with bias and uncertainty** + Data **with noise** = DA

- Also known as **filtering or state estimation**
- Goal: obtain the best statistical estimate of a natural system from partial observations of the true signal from nature

1. Prediction (Forecast)



2. Analysis (Filtering)



History of Data Assimilation

Core Idea: Data Assimilation (DA) combines:

- Imperfect mathematical models
- Noisy, sparse observations

to estimate the evolving state of a dynamical system.

Origins

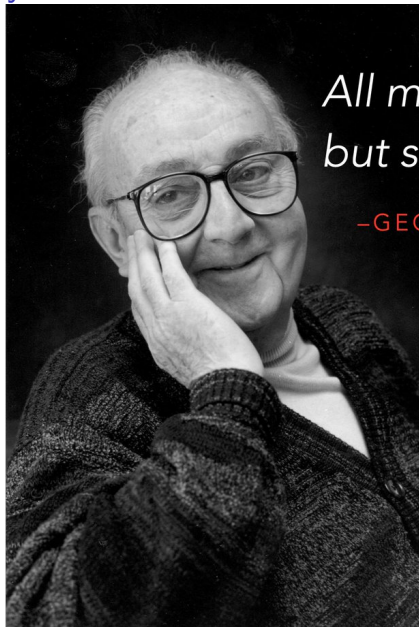
- 1950s–60s: Birth in meteorology
- 1960: Kalman Filter introduced (control theory)
- 1980s–90s: Variational methods (3DVAR / 4DVAR)
- 1990s–2000s: Ensemble methods (EnKF)

Today, DA is a **core component of numerical weather prediction**.

Operational milestone:

ECMWF Integrated Forecast System introduced 4D-Var in 1997.

Why Data Assimilation?: All Models Are Wrong



*All models are wrong,
but some are useful.*

—GEORGE BOX, UW-MADISON

Why Data Assimilation?

Data Assimilation solves:

- State estimation
- Parameter inference
- Model calibration

Mathematical View

DA is essentially:

Bayesian updating of dynamical systems

used to improve forecasts in large-scale nonlinear systems.

Applications of Data Assimilation

Atmosphere & Climate

- Satellite wind assimilation improves hurricane forecasting
- Used in global forecasting systems (e.g. ECMWF)

Oceanography

- Assimilation of SST and Argo floats
- Improves global ocean forecasts, sea ice dynamics

Engineering & Industry

- Oil reservoir history matching via EnKF
- Coastal current prediction for vessel traffic safety

Disaster Prediction

- Wildfire spread modeling using EnKF

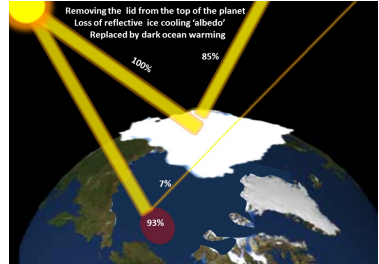
Modern Trend

DA now integrates with AI and deep learning for high-resolution forecasting.

Example 1: Climate Modeling

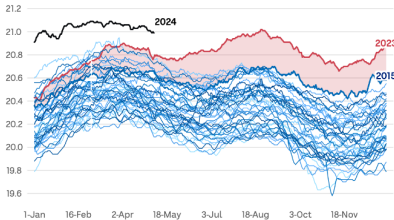
- Global warming, SST, heatwaves, wildfires, diseases
- Sea ice reflects 85% of the sunlight; ocean absorbs 93%
- Sea level rise, extreme weather, crop yield instability

Antarctic Region



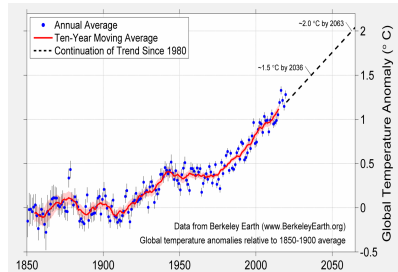
Credit: Paul Homewood

Daily mean sea surface temperature



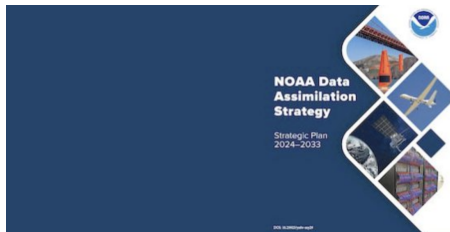
Each of these lines shows the daily global sea surface temperature across a year from 1979.

Source: [C3S Climate Data Store](#) / [Get the data](#)



Credit: Berkeley Earth

Example 1: Climate Modeling–NOAA



NOAA Releases 10-Year Strategy for Data Assimilation

📅 February 26, 2025

Co-Authors:

Kevin Garrett, NOAA/NWS

NOAA Data Assimilation Strategy Tiger Team Lead

Sergey Frolov, NOAA/OAR

NOAA Data Assimilation Strategy Tiger Team Co-Lead

Example 1: Climate Modeling–NOAA

1. Build the required infrastructure for continuous development and integration of data assimilation workflows. NOAA will rethink how it deploys and operates the end-to-end data assimilation process, toward continually updating the Earth-system state.
2. Maximize the use, value, and impact of Earth system observations. NOAA will leverage new techniques such as AI to extract, to the maximum extent, all information content from available environmental observations to inform Earth system analyses.
3. Transform and build the next-generation DA workforce with new skills beyond traditional DA expertise. This includes data scientists, software and cloud engineers, and experts on new sensor technologies. NOAA will leverage public and private partnerships to address these skill gaps.
4. Optimize organizational efficiencies across the research, development, and operational transition of new capabilities across the DA enterprise. NOAA will improve the alignment of research and operational priorities and work more seamlessly to integrate new observations and new techniques into our DA systems.

Example 2: Spacecraft



SpaceX CEO Elon Musk (Source: Dan G/Flickr)

Kalman Filtering for Spacecraft Attitude Estimation

E.J. Lefferts, F.L. Markley and M.D. Shuster

Published Online: 23 May 2012 • <https://doi.org/10.2514/3.56190>

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VOL. 5, NO. 5

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AIAA 82-0678R

Kalman Filtering for Spacecraft Attitude Estimation

E.J. Lefferts

NASA Goddard Space Flight Center, Greenbelt, Maryland

F.L. Markley

Naval Research Laboratory, Washington, D.C.

and

M.D. Shuster

Business and Technological Systems, Inc., Seabrook, Maryland

1. Introduction

THIS report reviews the methods of Kalman filtering in attitude estimation and their development over the last two decades. This review is not intended to be complete but is limited to algorithms suitable for spacecraft equipped with three-axis gyros as well as attitude sensors. These are the systems to which we feel that Kalman filtering is most applicable.

due to the use of rotating or rastering instruments lead to significant uncertainties in the modeling. For autonomous spacecraft the use of inertial reference units as a model replacement permits the circumvention of these problems. In this representation the angular velocity of the spacecraft is obtained from the gyro data. The kinematic equations are used to obtain the attitude state and this is augmented by means of additional state-vector components for the gyro

Example 3: AI/ML+DA

nature machine intelligence

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Article | Published: 22 July 2024

Partial-convolution-implemented generative adversarial network for global oceanic data assimilation

[Yoo-Geun Ham](#) , [Yong-Sik Joo](#), [Jeong-Hwan Kim](#) & [Jeong-Gil Lee](#)

[Nature Machine Intelligence](#) **6**, 834–843 (2024) | [Cite this article](#)

2528 Accesses | **8** Citations | **4** Altmetric | [Metrics](#)

 A [preprint version](#) of the article is available at Research Square.

Abstract



Journal of Computational Science

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Data Learning: Integrating Data Assimilation and Machine Learning ☆

[Caterina Buizza](#)^b, [César Quilodrán Casas](#)^a, [Philip Nadler](#)^a, [Julian Mack](#)^a, [Stefano Marrone](#)^{a,f},
[Zainab Titus](#)^c, [Clémence Le Cornec](#)^d, [Evelyn Heylen](#)^e, [Tolga Dur](#)^a, [Luis Baca Ruiz](#)^{a,g}, [Claire](#)
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Received 19 February 2020, Revised 8 October 2021, Accepted 2 December 2021, Available online 23 December 2021, Version of Record 5 January 2022.

Recommended Textbooks

- Majda, Andrew J., and John Harlim. Filtering complex turbulent systems. Cambridge University Press, 2012.
- Chen, Nan. Stochastic methods for modeling and predicting complex dynamical systems. Berlin/Heidelberg, Germany: Springer International Publishing, 2023.
- Law, Kody, Andrew Stuart, and Kostas Zygalakis. "Data assimilation." Cham, Switzerland: Springer 214 (2015): 52.
- Asch, Mark, Marc Bocquet, and Maëlle Nodet. Data assimilation: methods, algorithms, and applications. Society for Industrial and Applied Mathematics, 2016.

Tentative Schedule and Office Hours

1. Probability and Stochastic Toolkits
2. Information Theory
3. Stochastic Computational Methods
4. Kalman Filter and Its Variations
5. Conditional Gaussian Nonlinear Systems
6. Multiscale DA and Other DA Methods

Office Hours:

Thursdays 4pm-5pm

Shuangqing Complex Bldg. B523

Grading

Att./Quiz	12%	1% for each week; 12 out of 15 weeks
HW1	14%	Coding and theory, typed submission
HW2	14%	Coding and theory, typed submission
Project	50%	DA project, algorithm/method design, analysis, coding, and report, typed submission
Presentation	10%	Time/venue TBD

Motivation from Numerical Analysis

Deterministic Example: Improved Accuracy

Model problem: $y'(t) = 2t$, $y(0) = 0$, $y(t) = t^2$.

Model-only forecast (Forward Euler):

$$Y^{n+1} = Y^n + 2t_n \Delta t \quad \Rightarrow \quad Y^n = n(n-1)(\Delta t)^2, \quad \Rightarrow \quad e_Y^n = n(\Delta t)^2.$$

DA (analysis step):

$$Z^n = y(t_n) + \beta(\Delta t)^2, \quad U^n = \gamma Y^n + (1 - \gamma)Z^n.$$

Optimized blending:

$$\gamma^* = \frac{\beta}{\beta + (\Delta t)^2} \quad \Rightarrow \quad e_U^n = 0.$$

	Model-only	With DA (Optimal γ)
Global error	$\mathcal{O}(\Delta t)$	0 (perfect tracking)
RMSE	$\sim \Delta t$	$\sim \Delta t^2$ or 0
PCC	$\rightarrow 1$ slowly	$\rightarrow 1$ rapidly
Bias	accumulates	corrected each step

DA significantly improves accuracy and reduces bias.

Deterministic Example: Enhanced Stability

Model: $y'(t) = \lambda y$, $y(0) = 1$, $y(t) = e^{\lambda t}$.

Euler forecast: $Y^{n+1} = (1 + \lambda \Delta t) Y^n = b Y^n$, unstable if $|b| > 1$.

Stable if $0 < \Delta t < \frac{-2}{\lambda}$

DA-cycled scheme:

$$U^{n+1} = \gamma b U^n + (1 - \gamma)(a^{n+1} + \beta), \quad a = e^{\lambda \Delta t}.$$

Error dynamics:

$$e^{n+1} = \gamma b e^n + \gamma(a - b)a^n - (1 - \gamma)\beta.$$

Stability condition:

$|\gamma b| < 1 \Rightarrow$ DA stabilizes even when model-only forecast diverges.

DA anchors forecasts to observations, ensuring stability and bounded error.

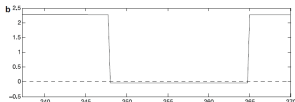
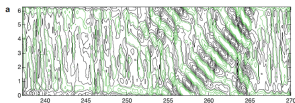
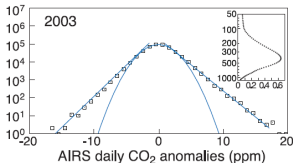
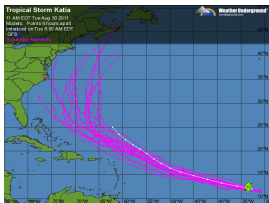
Probability and Stochastic Toolkit

Main reference

C. W. Gardiner, Stochastic methods: A handbook for Natural and Social Sciences, *Springer*, 2010.

B. Oksendal, Stochastic differential equations, *Springer*, 2000, 5ed.

Motivations



1. If we know the *stochastic* model ...

- ▶ how to solve the path-wise solution?
- ▶ how to solve the probability distribution function (PDF)?

2. Gaussian v.s. non-Gaussian systems ...

- ▶ Features of Gaussian and non-Gaussian distributions.
- ▶ Solving the time-dependent moments/PDF (linear system v.s. nonlinear system).

3. Building complex model with regime switching ...

- ▶ Hidden variables.
- ▶ Intermittency.

Basic concepts of probability.

1. Random variable.

A random variable $\mathbf{X}: \Omega \rightarrow E$ is a measurable function from the set of possible outcomes Ω to some set E . Usually, $E = \mathbb{R}^n$.

2. Probability density function (PDF).

If there exists a nonnegative, integrable function $p: \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$F_{\mathbf{X}}(\mathbf{x}) = P(\mathbf{X} \leq \mathbf{x}) = \int_{-\infty}^{x^1} \cdots \int_{-\infty}^{x^n} p(y^1, \dots, y^n) dy^1 \dots dy^n,$$

where $\mathbf{x} = (x^1, \dots, x^n)$, then p is called the probability density function for $\mathbf{X}$.

3. Marginal distribution.

Suppose that $\mathbf{X}: \Omega \rightarrow \mathbb{R}^n$ is a random variable with PDF $p(x_1, \dots, x_n)$. The *marginal probability density* $\tilde{p}(x_i)$ for x_i , $i \in [1, \dots, n]$ is given by

$$\tilde{p}(x_i) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} p(y_1, \dots, x_i, \dots, y_n) dy_1 \dots dy_{i-1} dy_{i+1} \dots dy_n.$$

4. Moments.

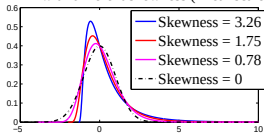
Consider 1-D case for simplicity. The n -th moment and central moment of X are defined respectively by

$$\mu^n = E[X^n] = \int_{-\infty}^{\infty} x^n p(x) dx, \quad n \geq 1,$$

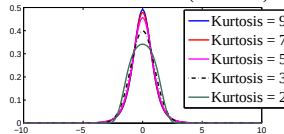
$$\tilde{\mu}^n = E[(X - \mu)^n] = \int_{-\infty}^{\infty} (x - \mu)^n p(x) dx, \quad n \geq 2.$$

Statistics	Description	Moment	Formula
Mean	the central tendency	1-st moment	μ
Variance	spreading out	2-nd central moment	$\sigma^2 = \tilde{\mu}^2$
Skewness	the asymmetry	3-rd normalized central moment	$\tilde{\mu}^3 / \sigma^3$
Kurtosis	the “peakedness”	4-th normalized central moment	$\tilde{\mu}^4 / \sigma^4$

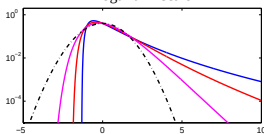
PDF with different skewness (linear scale)



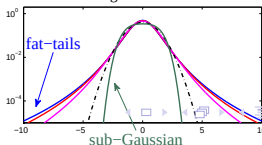
PDF with different kurtosis (linear scale)



Logarithm scale



Logarithm scale



5. Gaussian distribution.

Gaussian distribution $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$,

$$p(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}.$$

The skewness of the Gaussian distribution is 0 and kurtosis is 3.

For a Gaussian random variable, the **marginal density** over a subset of components is obtained by simply **dropping the remaining components**, e.g. given a Gaussian vector $(x_1, x_2, x_3)^T$ with $\boldsymbol{\mu} = (\mu_1, \mu_2, \mu_3)^T$ and covariance

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} \\ \Sigma_{21} & \Sigma_{22} & \Sigma_{23} \\ \Sigma_{31} & \Sigma_{32} & \Sigma_{33} \end{pmatrix},$$

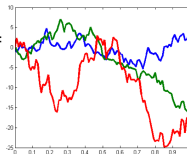
the marginal density for (x_1, x_3) is given by a Gaussian with

$$\tilde{\boldsymbol{\mu}} = \begin{pmatrix} \mu_1 \\ \mu_3 \end{pmatrix}, \quad \tilde{\Sigma} = \begin{pmatrix} \Sigma_{11} & \Sigma_{13} \\ \Sigma_{31} & \Sigma_{33} \end{pmatrix}.$$

Introduction to Ito stochastic calculus and SDE's.

1. Stochastic process.

- ▶ A stochastic process is a collection of random variables $\{X_t(\omega) | t \in \mathbb{R}^+\}$, representing the evolution of some system of random values over *time*.
- ▶ For each sample point $\omega \in \Omega$, the mapping $t \rightarrow X_t(\omega)$ is the corresponding sample path of the stochastic process.



2. Wiener process (Brownian motion).

Real-valued stochastic process $W(t)$ such that

- ▶ $W(0) = 0$ a.s. (with probability one).
- ▶ $W(t)$ is continuous with probability one.
- ▶ $W(t)$ has independent increments with distribution $W(t) - W(s) \sim \mathcal{N}(0, t - s)$ for $0 < s < t$.

Properties: for each time $t \geq 0$,

- ▶ $E(W(t)) = 0$,

► $E(W^2(t)) = t.$

$$\begin{aligned}\Delta W_i &\equiv W(t_i) - W(t_{i-1}) \\ E(\Delta W_i^2) &= \Delta t_i\end{aligned}$$

Sample paths of the Wiener process are, with probability one, nowhere differentiable.

3. Ito stochastic differential equation.

Stochastic process X satisfies the Ito SDE

$$dX(t) = A(X(t), t)dt + B(X(t), t)dW(t), \quad t > t_0 > 0, X(t_0) = X_0$$

if for all t and t_0

$$X(t) = X(t_0) + \underbrace{\int_{t_0}^t A(X(s), s)ds}_{\text{Stieltjes integral}} + \underbrace{\int_{t_0}^t B(X(s), s)dW(s)}_{\text{Ito integral}}.$$

- ▶ $X(t)$ is required to be adapted (non-anticipating)¹.
- ▶ Ito stochastic integral is defined as

$$\int_{t_0}^t B(X(s), s)dW(s) := m.s. \lim_{n \rightarrow \infty} \sum_{j=1}^n B(X(t_{j-1}), t_{j-1})(W(t_j) - W(t_{j-1})),$$

where $m.s. \lim_{n \rightarrow \infty} X_n = X \rightarrow \lim_{n \rightarrow \infty} \langle (X_n - X)^2 \rangle = 0$.

Basic properties: For non-anticipating function G, H ,

- ▶ $\int_{t_0}^t G(t')[dW(t')]^2 = \int_{t_0}^t G(t')dt', \quad (dW^2(t) = dt),$
- ▶ $\langle \int_{t_0}^t G(t')dW(t') \rangle = 0,$
- ▶ $\langle \int_{t_0}^t G(t')dW(t') \int_{t_0}^t H(t')dW(t') \rangle = \int_{t_0}^t \langle G(t')H(t') \rangle dt.$

¹ $G(t)$ is nonanticipating if $G(t)$ is statistically independent of the increments $W(s) - W(t)$ for all t and s with $t < s$.

4. Ito formula.

Assume X_t satisfies

$$dX_t = a_t dt + b_t dW_t,$$

where $a_t := a(X(t), t)$, $b_t := b(X(t), t)$. Then for a smooth deterministic function f , we have

4. Ito formula.

Assume X_t satisfies

$$dX_t = a_t dt + b_t dW_t,$$

where $a_t := a(X(t), t)$, $b_t := b(X(t), t)$. Then for a smooth deterministic function f , we have

$$\begin{aligned} df(X_t) &= f'(X_t)dX_t + \frac{1}{2}f''(X_t)(dX_t)^2 \\ &= f'(X_t)(a_t dt + b_t dW_t) + \frac{1}{2}f''(X_t)(a_t dt + \underline{b_t dW_t})^2 \\ &= f'(X_t)(a_t dt + b_t dW_t) + \frac{1}{2}f''(X_t)b_t^2 dt \\ &= \left(f'(X_t)a_t + \frac{1}{2}f''(X_t)b_t^2\right)dt + f'(X_t)b_t dW_t, \end{aligned}$$

where we have used the following fact

$$dW_t \sim \mathcal{N}(0, dt) \implies dW_t \cdot dW_t = dt \text{ (variance).}$$

5. Fokker-Planck equation.

The Fokker-Planck equation describes the time evolution of the PDF $p(x, t)$ associated with a stochastic process x ,

$$dx = a(x, t)dt + b(x, t)dW_t.$$

Assume $f(x(t))$ is a smooth function. Then its mean evolution satisfies

$$\begin{aligned}\frac{d}{dt}\langle f(x(t)) \rangle &= \left\langle a(x(t), t) \frac{\partial f(x)}{\partial x} + \frac{1}{2} b^2(x(t), t) \frac{\partial^2 f(x)}{\partial x^2} \right\rangle && \text{(Ito formula)} \\ &= \int_{-\infty}^{\infty} \left(a(x(t), t) \frac{\partial f(x)}{\partial x} + \frac{1}{2} b^2(x(t), t) \frac{\partial^2 f(x)}{\partial x^2} \right) p(x, t) dx \\ &= \int_{-\infty}^{\infty} f(x) \left(-\frac{\partial}{\partial x} (a(x(t), t)p(x, t)) + \frac{1}{2} \frac{\partial^2}{\partial x^2} (b^2(x(t), t)p(x, t)) \right) dx.\end{aligned}$$

On the other hand

$$\frac{d}{dt}\langle f(x(t)) \rangle = \int_{-\infty}^{\infty} f(x) \frac{\partial p(x, t)}{\partial t} dx.$$

Thus, the Fokker-Planck equation reads

$$\frac{\partial p(x, t)}{\partial t} = \underbrace{-\frac{\partial}{\partial x} (a(x(t), t)p(x, t))}_{\text{drift}} + \underbrace{\frac{1}{2} \frac{\partial^2}{\partial x^2} (b^2(x(t), t)p(x, t))}_{\text{Diffusion}}.$$

Scalar SDE: $dx = a(x, t)dt + b(x, t)dW_t$,

F-P equation: $\frac{\partial p(x, t)}{\partial t} = -\frac{\partial}{\partial x} \left(a(x(t), t) p(x, t) \right) + \frac{1}{2} \frac{\partial^2}{\partial x^2} \left(b^2(x(t), t) p(x, t) \right).$

Simple example in scalar case: Assume initial condition $p(x, 0) = \delta(x)$.

$$\begin{aligned} dx &= dt \\ \Rightarrow \frac{\partial p(x, t)}{\partial t} &= -\frac{\partial p(x, t)}{\partial x} \\ \Rightarrow p(x, t) &= \delta(x - t). \end{aligned}$$

$$\begin{aligned} dx &= dW_t \\ \Rightarrow \frac{\partial p(x, t)}{\partial t} &= \frac{1}{2} \frac{\partial^2 p(x, t)}{\partial x^2} \\ \Rightarrow p(x, t) &= \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}. \end{aligned}$$

$$d\mathbf{x} = \mu(\mathbf{x}, t)dt + \sigma(\mathbf{x}, t)d\mathbf{W}_t,$$

where $\sigma(\mathbf{x}, t)$ is an $N \times M$ matrix and $\mathbf{W}_t$ is an M -dimensional Wiener process.

The associated Fokker-Planck equation reads

$$\frac{\partial p(\mathbf{x}, t)}{\partial t} = - \sum_{i=1}^N \frac{\partial}{\partial x_i} [\mu_i(\mathbf{x}, t) p(\mathbf{x}, t)] + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2}{\partial x_i \partial x_j} [D_{ij}(\mathbf{x}, t) p(\mathbf{x}, t)]$$

with drift vector $\mu = (\mu_1, \dots, \mu_N)$ and diffusion tensor

$$D_{ij}(\mathbf{x}, t) = \sum_{k=1}^M \sigma_{ik}(\mathbf{x}, t) \sigma_{jk}(\mathbf{x}, t).$$

► Scalar case:

Scalar SDE: $dx = a(x, t)dt + b(x, t)dW_t$,

F-P equation:
$$\frac{\partial p(x, t)}{\partial t} = -\frac{\partial}{\partial x} \left(a(x(t), t)p(x, t) \right) + \frac{1}{2} \frac{\partial^2}{\partial x^2} \left(b^2(x(t), t)p(x, t) \right).$$

Simple example in scalar case: Assume initial condition $p(x, 0) = \delta(x)$.

$dx = dt$ $\Rightarrow \frac{\partial p(x, t)}{\partial t} = -\frac{\partial p(x, t)}{\partial x}$ $\Rightarrow p(x, t) = \delta(x - t).$	$dx = dW_t$ $\Rightarrow \frac{\partial p(x, t)}{\partial t} = \frac{1}{2} \frac{\partial^2 p(x, t)}{\partial x^2}$ $\Rightarrow p(x, t) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}.$
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► Consider a n -dimensional process $\mathbf{x}$,

$$d\mathbf{x} = \boldsymbol{\mu}(\mathbf{x}, t)dt + \boldsymbol{\sigma}(\mathbf{x}, t)d\mathbf{W}_t,$$

where $\boldsymbol{\sigma}(\mathbf{x}, t)$ is an $N \times M$ matrix and $\mathbf{W}_t$ is an M -dimensional Wiener process.

The associated Fokker-Planck equation reads

$$\frac{\partial p(\mathbf{x}, t)}{\partial t} = -\sum_{i=1}^N \frac{\partial}{\partial x_i} [\mu_i(\mathbf{x}, t)p(\mathbf{x}, t)] + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2}{\partial x_i \partial x_j} [D_{ij}(\mathbf{x}, t)p(\mathbf{x}, t)]$$

with drift vector $\boldsymbol{\mu} = (\mu_1, \dots, \mu_N)$ and diffusion tensor

$$D_{ij}(\mathbf{x}, t) = \sum_{k=1}^M \sigma_{ik}(\mathbf{x}, t)\sigma_{jk}(\mathbf{x}, t).$$

However ... solving F-P equation is impractical for high dimensional systems.

Practical strategies will be shown in the following lectures.

Summary

- ▶ A stochastic process is a collection of random variables, representing the evolution of some system of random values over *time*.
- ▶ A **Brownian motion** $W(t)$ is continuous but nowhere differentiable and has independent increments with distribution $W(t) - W(s) \sim \mathcal{N}(0, t - s)$.
- ▶ **Left endpoint** is taken in the Ito integral.
- ▶ **Ito formula**. Assume x_t satisfies

$$dx_t = a_t dt + b_t dW_t,$$

Then for a smooth deterministic function f , we have

$$df(x_t) = \left(f'(x_t)a_t + \frac{1}{2}f''(x_t)b_t^2 \right) dt + f'(x_t)b_t dW_t.$$

- ▶ The **Fokker-Planck equation** describes the time evolution of the PDF $p(x, t)$ associated with a stochastic process.

$$\frac{\partial p(x, t)}{\partial t} = \underbrace{-\frac{\partial}{\partial x} \left(a(x(t), t)p(x, t) \right)}_{\text{drift}} + \underbrace{\frac{1}{2} \frac{\partial^2}{\partial x^2} \left(b^2(x(t), t)p(x, t) \right)}_{\text{Diffusion}}.$$

Thank you! Questions? Comments?